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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

AD2307-1 – INTERNET OF THINGS

(For AD)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following best describes the Internet of Things (IoT)?
A) A network of interconnected computers B) A network of interconnected devices embedded with software, sensors, and network connectivity, which enables data exchange.
C) A network of interconnected mobile phones D) A network of interconnected servers
2. The genesis of IoT can be traced back to:
A) The invention of the internet B) The rise of social media
C) The development of microprocessors and wireless communication D) The invention of the personal computer
3. How does IoT relate to digitization?
A) IoT is a subset of digitization B) IoT and digitization are unrelated concepts.
C) Digitization is a subset of IoT D) IoT is a key enabler of digitization by providing data and connectivity
4. Which of the following is not a key challenge in IoT?
A) Security and privacy B) Scalability
C) Interoperability D) Size
5. Which layer is NOT typically part of a simplified IoT architecture?
A) Application Layer B) Network Layer
C) Presentation Layer D) Device Layer
6. Which of the following is NOT an IoT access technology?
A) SQL B) Wi-Fi
C) Bluetooth D) Ethernet
7. What is the primary security concern in IoT networks?
A) Data encryption B) Device interoperability
C) Network congestion D) Cloud storage availability
8. What operating system is commonly used with Raspberry Pi?
A) Windows 10 B) Ubuntu Server
C) Raspbian D) MacOS
9. Which of the following interfaces is NOT commonly found on a Raspberry Pi?
A) HDMI B) USB
C) GPIO D) PCIe
10. Which command is used to update installed packages on Raspberry Pi's Linux OS?
A) sudo upgrade B) sudo apt-get update
C) sudo refresh D) sudo install update

11. Which Python library is commonly used for controlling GPIO pins on a Raspberry Pi?
 A) RPi.GPIO B) gpioPython
 C) PiGPIO D) RaspberryIO
12. Which of the following is used to set up a pin as an input on Raspberry Pi using Python?
 A) GPIO.setup(pin, GPIO.OUT) B) GPIO.setup(pin, GPIO.IN)
 C) GPIO.input(pin, GPIO.HIGH) D) GPIO.output(pin, GPIO.LOW)
13. Which microcontroller is commonly used in Arduino boards?
 A) ARM Cortex-M4 B) ATmega328P
 C) ESP8266 D) STM32
14. Which function is used to read input from a digital sensor in Arduino?
 A) digitalRead(pin) B) digitalWrite(pin)
 C) analogRead(pin) D) pinMode(pin)
15. Which component is used to drive a DC motor in an Arduino project?
 A) LED B) Resistor
 C) Capacitor D) Relay
16. Which module is used for temperature and humidity sensing in Raspberry Pi projects?
 A) MPU6050 B) DHT11
 C) HC-SR04 D) LDR
17. What happens when you apply a HIGH signal to an LED connected to Raspberry Pi GPIO?
 A) The LED turns off B) The LED blinks rapidly
 C) The LED turns on D) The LED gets damaged
18. Ultrasonic sensor is used for
 A) Measuring the distance B) Measuring the sound
 C) Measuring the temperature D) Measuring the light voltage
19. In a constant resistant voltage is proportional to
 A) Current B) Power
 C) Size of the device D) Time
20. Identify the odd man out
 A) Nano B) Uno
 C) Pi D) Mega

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the IoT Functional Stack with suitable diagram.	8	L*2	1	1
	b) Consider the concept of a smart city. Apply the IoT concept to this and explain any five devices that can be developed for smart city.	8	L3	2	1
2.	a) Apply the IoT Level – 3 to a smart farming application and explain the architecture and components.	8	L3	2	1
	b) Define IoT and explain its characteristics.	8	L2	1	1
3.	a) List any four differences between M2M and IoT	8	L1	1	1
	b) Write a purpose, requirement, and process specification for smart parking application.	8	L3	2	1
Unit – II					
4.	a) Develop a program and circuit diagram to demonstrate the use of an IR sensor for counting the customers in a shop.	8	L3	3	1
	b) With a suitable diagram, Explain the parts of an Arduino Uno board. How is it different from a Raspberry Pi?	8	L2	4	1
5.	a) With a suitable example program and circuit, demonstrate how the moving objects are detected using the PIR sensor and can be used for automated light control.	8	L3	3	1

- b) What is a CSV file? Explain. Develop a python program to read ten different data from the user and write it to a CSV file. Read the data from the file and print it on the screen.

8 L3 3 1

6. a) Explain the structure and pins of the Raspberry Pi 3B model with the help of a diagram.

8 L2 5 1

- b) Design an Arduino program and circuit to demonstrate the working of a push button to toggle the working of an LED between on and off state.

8 L3 4 1

Unit – III

7. a) Explain the process of sending data to the cloud from Raspberry Pi

8 L2 5 1

- b) Demonstrate the process of sending data from Arduino to a raspberry pi.

8 L3 4 1

8. a) Develop a program in python to Blink LED and make buzzer sound alternatively. Show the circuit diagram for the same.

8 L3 3 1

- b) Develop a program and circuit to start a DC motor (fan) when a threshold temperature is reached in DHT11, using Raspberry Pi. Stop the fan when a push button is pressed.

8 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
